AI and the Future of Instructional Design in Education: Analyzing Job Market Trends and Skills

Analytics Practice Assignment

Allison Hoekstra, Rebecca Yost, and Stephen Lindhorst

Florida State University

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Dr. Bret Staudt Willet

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Table of Contents

Identification of Problem or Opportunity 3
Data Collection Tool and Process Description 4
Collecting Initial ID Job Descriptions Dataset 4
Cleaning and Processing ID Job Descriptions 4
NLTK Analysis of ID Job Descriptions 5
Separate Analysis of Faculty & Director Positions 5
Output Analysis 6
ID Job Skills Overview 6
ID AI and Emerging Tech Skills 8
Role Types 8
AI and Emerging Tech Skills 9
ID Education and Experience Requirements 12
Faculty & Director – AI and Emerging Tech Skills Analysis 12
Strengths and Limitations 15
Visualizations Graveyard – Rest in peace 17
References 20

Identification of Problem or Opportunity

Is AI going to take over my job as an instructional designer? Look at any recent news article and you'll be treated to chaotic predictions surrounding AI and its impact on the workplace. In September 2024, a mix of over 500 instructional design professionals were surveyed about their use of AI: about 30% used AI weekly, with 50% stating they used AI daily (Alster, 2025).

Instructional designers play an important role in leading and shaping the adoption of such technologies in educational institutions, but it is still relatively unknown how instructional designers themselves are supposed to utilize AI on the job. Choi et al. (2024) state that only a few studies address the use of generative AI for designing instruction. Kumar et al. (2024) echo that there is scarce literature on the subject but provide some findings from Kim and Adlof (2024) regarding the "pedagogical benefits of LLMs like ChatGPT as productivity, preparation, and problem-solving." These early explorations give some insight into how instructional designers may be expected to use AI for instruction, but more data and analysis are needed to understand how the demands of the job market are being impacted by AI over time. This analysis uses instructional designer job description data to inform instructional designers and program managers in higher education of the expectations prospective employers may have.

Data Collection Tool and Process Description

Collecting Initial ID Job Descriptions Dataset

One month of instructional design job postings were collected from HigherEdJobs. Two job types were selected: Faculty and Admin *Instructional Technology & Design* positions. One month of jobs resulted in 144 job descriptions. Each job description from the site was printed to PDF and then merged into a comprehensive file for archival purposes. The merged PDF file also preserved the visual fidelity of the job description. A web-scraper was not used because of ethical concerns and the lack of standardized fields shared by job descriptions.

Cleaning and Processing ID Job Descriptions

To make the dataset more manageable, we focused on the 100 most recent job posts. Each post was reviewed to determine if the job included instructional design tasks. Two of our group manually reviewed postings to identify and remove unrelated and part-time positions, classify positions as staff or faculty, and identify which roles required AI skills. Many jobs cleaned were faculty positions in the field of instructional design, while others were IT focused, or temporary internships. 62 instructional design jobs were left after cleaning and were standardized for position, job type: faculty or staff, date, location, salary, educational requirements, experience, job skills, and AI skills. ChatGPT assisted with cleaning and normalizing values so that they shared the same formatting.

In addition to analyzing job posts for AI skills, a broader analysis of related terms was undertaken. In a study that emerged during the data collection period, Roberts and McCormick (2025) indicated that 48% of institutions are either primarily upskilling existing

staff or using a mix of upskilling and hiring; only 1% of institutions are primarily hiring new staff in AI-related roles. Thus, the cleaned dataset of 62 postings was first manually searched for mentions of "innovative," "emerging" or "current" technologies, “professional development,” and “trends,” as these terms could be reasonably construed as encompassing the ability to learn or use AI on the job. The dataset was uploaded into ChatGPT to identify additional job postings that met this criteria, were reviewed manually, and either retained for analysis or discarded. The search found an additional 13 positions requiring the ability to use or learn emerging technologies.

NLTK Analysis of ID Job Descriptions

A job skills analysis was done using NLTK in Python using code created by ChatGPT. For this analysis, descriptions of knowledge, skills and abilities were copied from the PDF by row into one column of a spreadsheet for each of the job descriptions. When prompted, ChatGPT created a Python script that combined all the text in the specified Excel spreadsheet column. Then NLTK analyzed the text for the most commonly occurring words and phrases. From this analysis, we could see the most mentioned job skills required for instructional designers in the dataset.

Separate Analysis of Faculty & Director Positions

As the 100 positions were cleaned and sorted, several of the faculty positions that were being removed mentioned AI skills. Therefore, to gain a deeper understanding of the relationship between higher education and AI skills, these faculty, associate director, and director positions within instructional design programs in higher education were analyzed separately. Job postings for instructional designers, staff, and technical roles were

excluded. After data cleaning, 20 relevant job postings remained. These roles were analyzed based on job type (faculty or director), program level (undergraduate or graduate and above), as well as AI and emerging technology skills.

Output Analysis

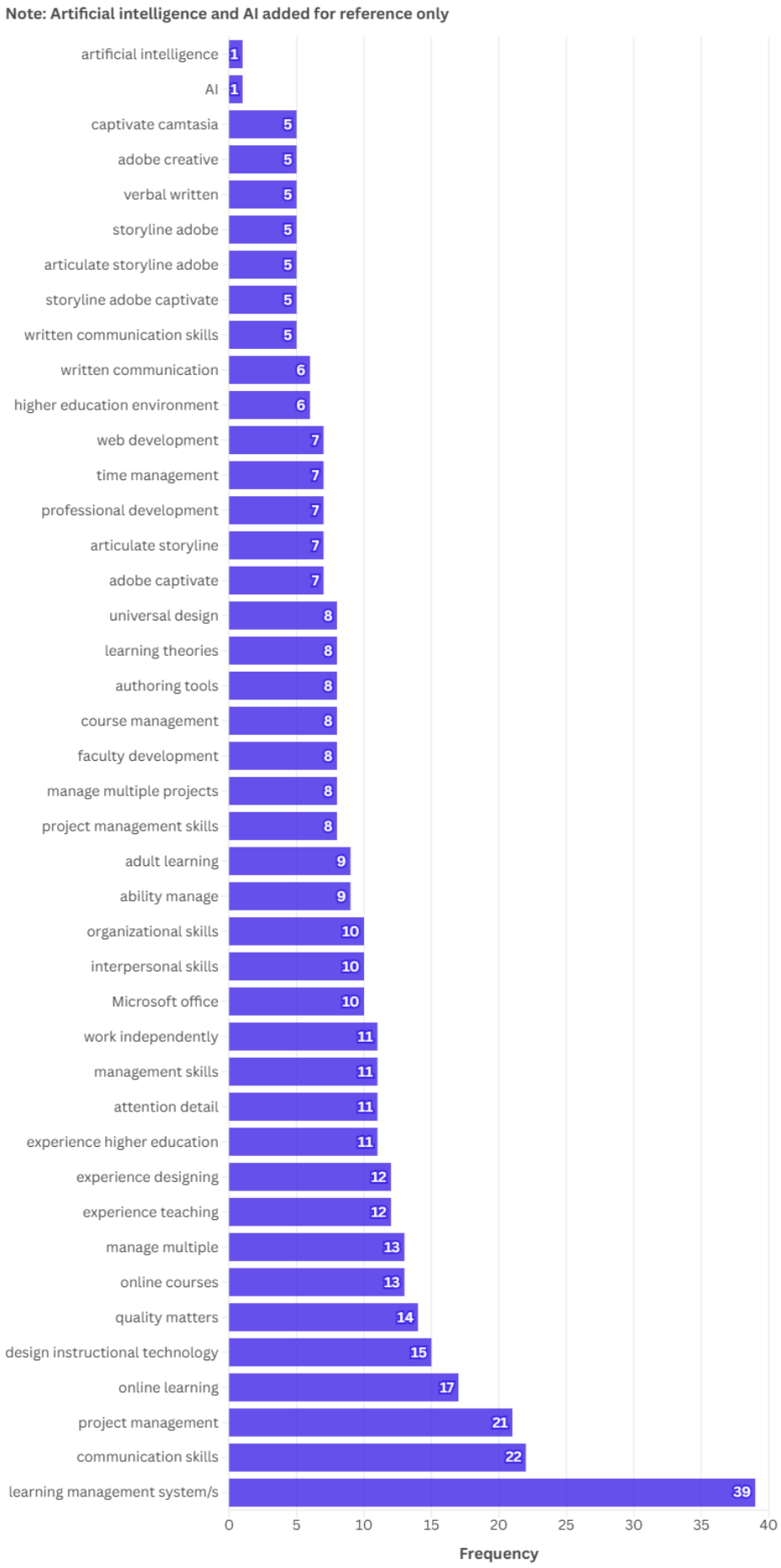
After cleaning the dataset, analyses were conducted to address the following questions: What job skills are required for IDs in higher education? How many, and what type of roles require AI skills? Is there a trend in either AI skills or emerging technologies requirements? And how many job postings for faculty required AI skills, compared instructional designers?

ID Job Skills Overview

Skill with learning management systems LMSs was the most frequently requested skill (39 times) by employers as seen below in Figure 1. It's important to note that out of 62 job descriptions, only 1 mention of *artificial intelligence* and *AI* was stated as a job requirement. However, there were more mentions of AI and emerging tech skills found outside of job requirements which are explored next.

Figure 1

The Most Frequent Instructional Design Job Skills Required



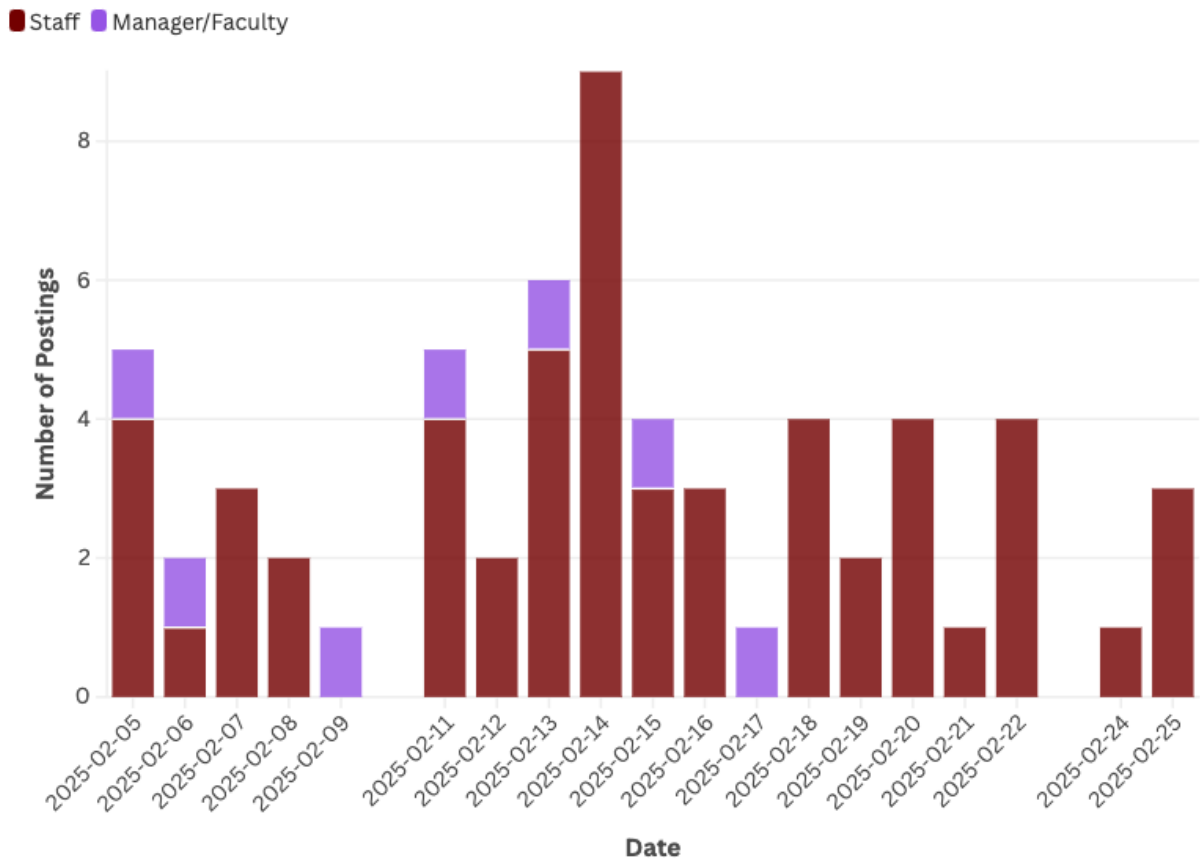
ID AI and Emerging Tech Skills

Role Types

To determine the overall number of postings available in various instructional design-related roles (staff versus faculty or managerial), a basic frequency count sufficed. The following visualization shows the number of positions posted broken out by role.

Figure 2

Distribution of Staff Positions Versus Faculty or Managerial Positions

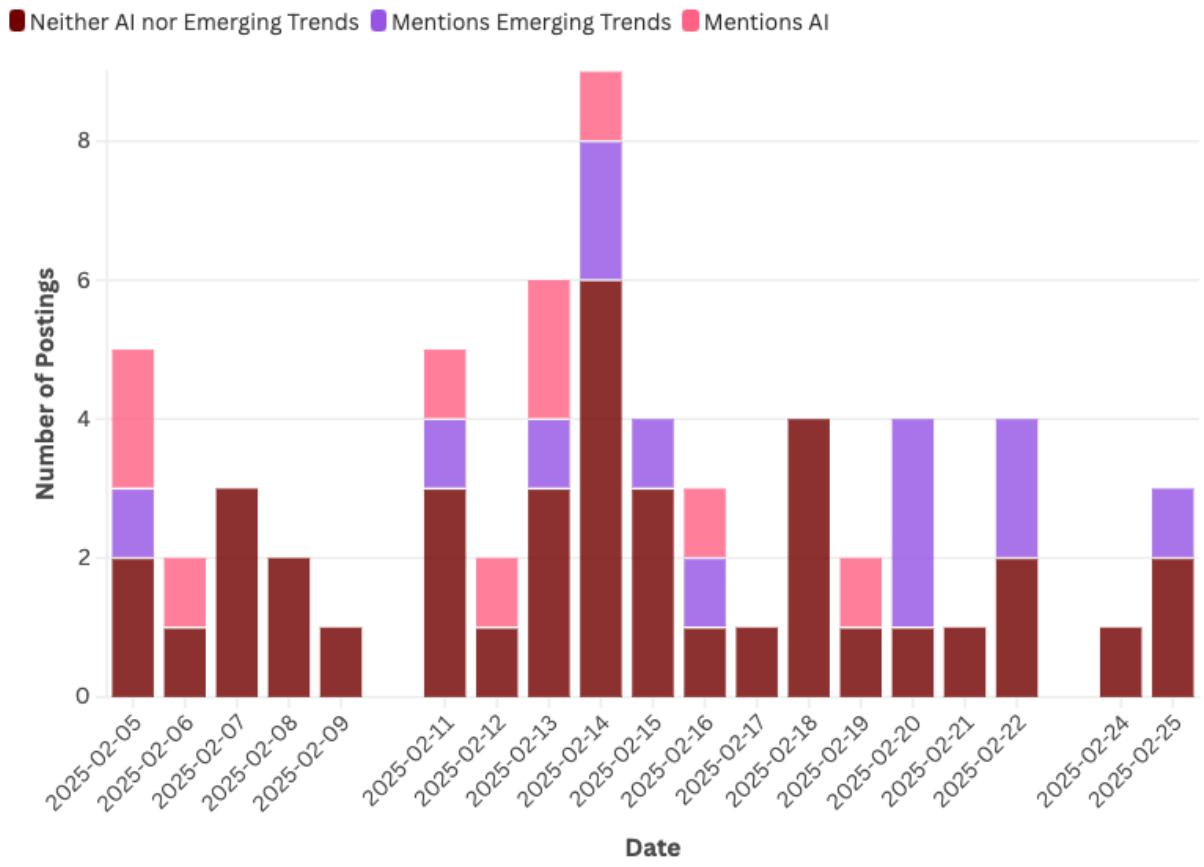


AI and Emerging Tech Skills

10 positions in the dataset included AI skills, and 13 additional positions required emerging technology skills. Figure 3 demonstrates the relative frequencies of each of these positions by day.

Figure 3

Number of AI and Emerging Trends Requirements in Postings

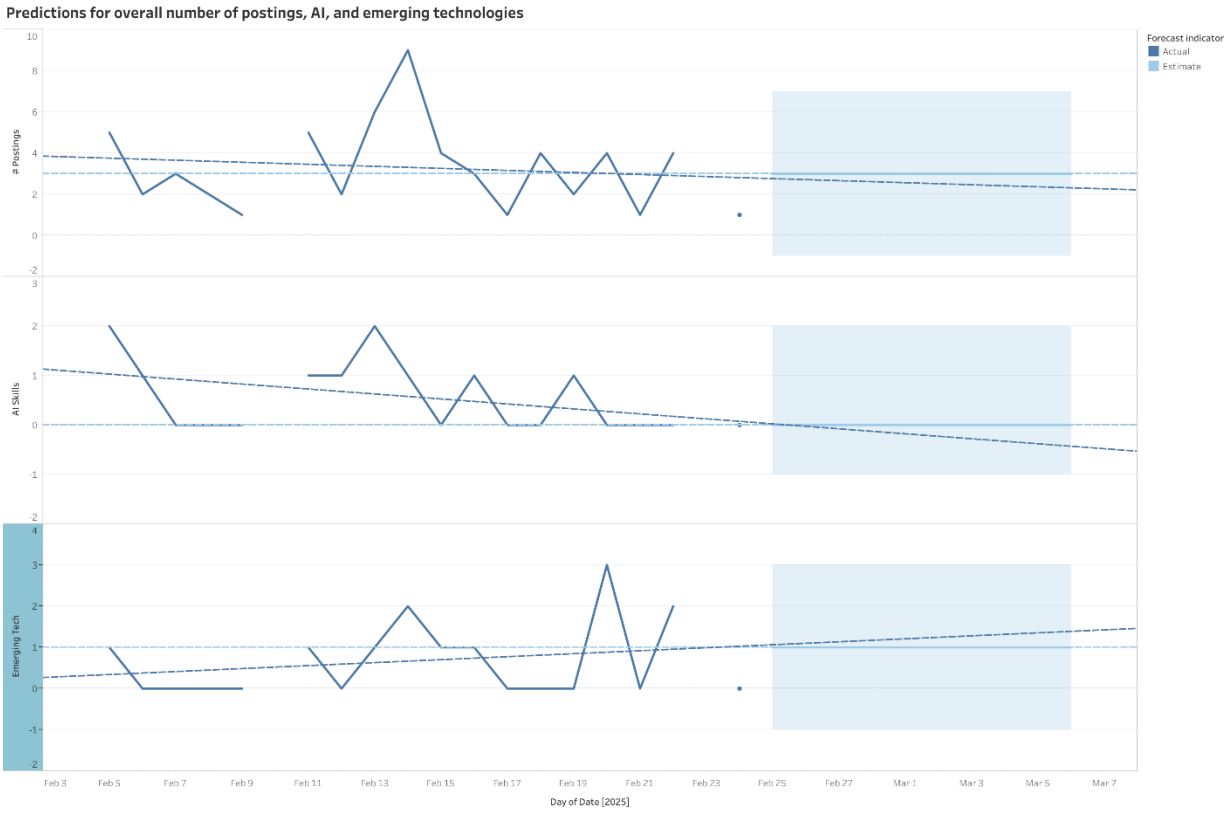


Based on our observations, 16% of postings requested AI skills and 21% expected KSAs in new technologies or trends. It might be reasonable to predict that these patterns would continue the following month. To determine whether this assumption is valid, trendlines were generated in Tableau to demonstrate a linear relationship in the number of

postings and postings containing AI skills over time. According to this analysis, it might be reasonable that between 0-7 positions could be posted per day in the coming weeks, but most likely 3 postings, and that fewer than 2 postings per day would request AI skills but would most likely post 0 positions. See Figure 4:

Figure 4

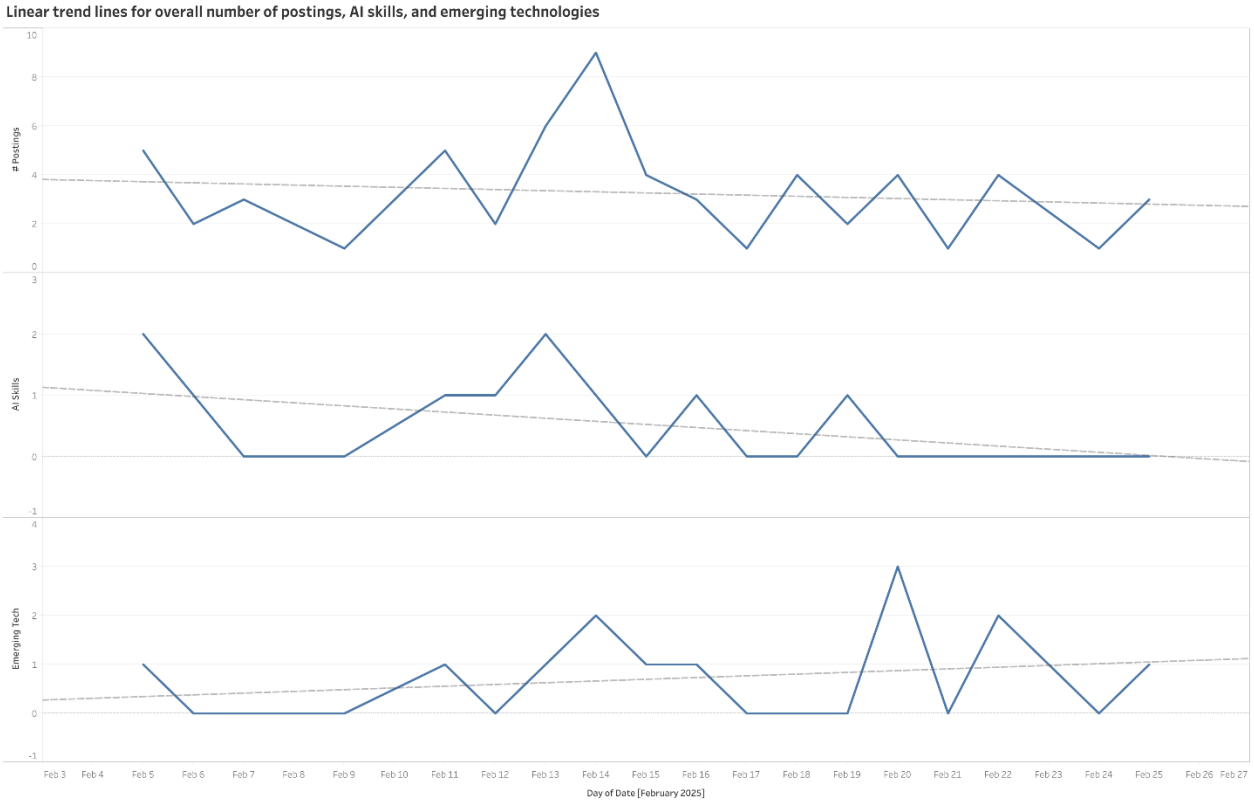
Predictions for Overall Number of Postings, AI, and Emerging Technologies



In general, there was a downward trendline in positions posted during the month of February. Additionally, there was a downward trend observed in positions requiring AI-related skills, but a slight upward trend in positions posted with “emerging” or “current” technologies. Figure 5 below shows the overall trends.

Figure 5

Linear Trend Lines for Overall Postings, AI skills, and Emerging Technologies



Relatively high p-values between .31 and .65 were found for most of these analyses, which suggests that these shifts could be due to random chance. The only analysis with a p-value of .05 was the downward trend in positions requiring AI related skills. With a longer observation period and a larger dataset this relationship could be further investigated and its predictive value assessed for gauging the impact of AI skills on instructional design job postings in higher education.

ID Education and Experience Requirements

For minimum education, most ID jobs required either a bachelor's (47%) or a master's degree (38%). For preferred requirements a master's (55%) and a doctorate (27%) were the most common. This demonstrates that a bachelor's is the standard minimum to enter the field, but the safest bet is likely a master's degree as nearly half of all minimum and preferred requirements specified a master's degree. With advanced degree requirements becoming the norm, this makes the role of faculty and directors of ID programs even more important.

Figure 6

Minimum Education Reqs for IDs

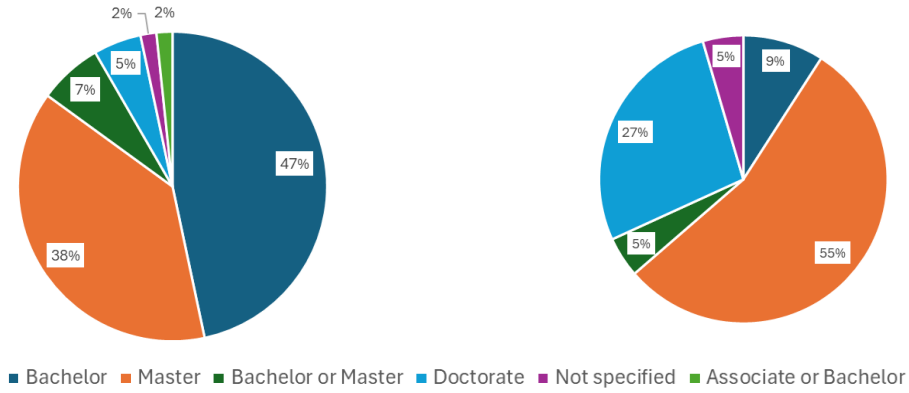


Figure 7

Preferred Education Reqs for IDs

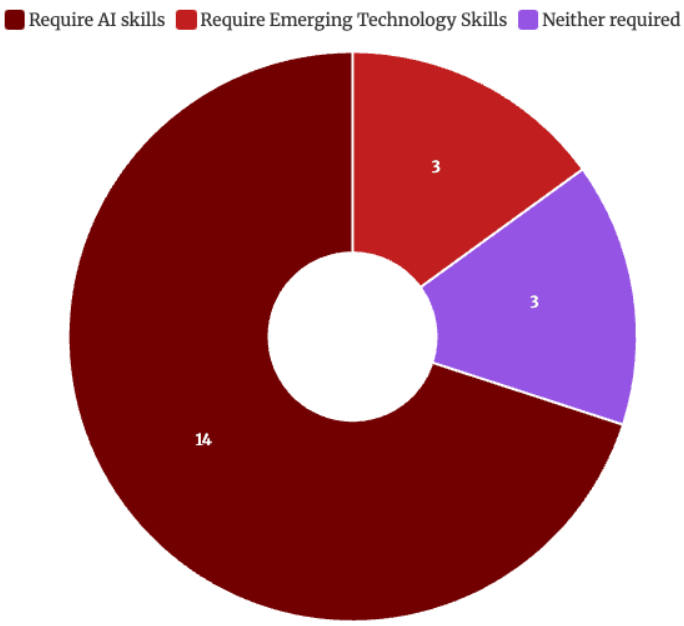
Faculty & Director – AI and Emerging Tech Skills Analysis

The analysis of the 20 job postings for faculty and director roles in instructional design programs in higher education indicated a strong desire for AI and emerging technology skills. Seventeen of the job postings required experience in AI or emerging technologies. Fourteen postings required AI-related skills, while three of the job postings required experience with emerging technologies or industry trends, phrasing requirements such as found in job posting #64, "Faculty are expected to stay current with industry trends

and integrate real-world applications into their teaching." Only three of the twenty positions did not list AI or emerging technology requirements.

Figure 8

Faculty & Director Role Required Skills

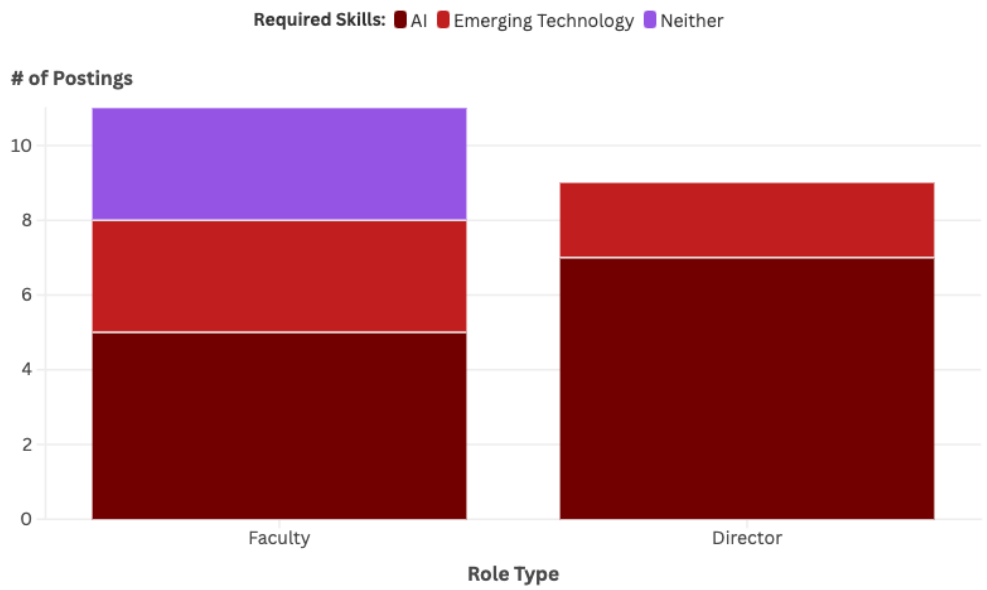


Notably, no director or split faculty/director job postings lacked AI or emerging technology requirements. Of the nine job postings for these roles seven explicitly required AI-related skill sets. These postings emphasized qualifications such as "experience with innovative pedagogy, such as the incorporation of AI into teaching," as stated in job posting #64. Job posting #13 and #81 focused on broader technological skill sets, requiring experience with "innovative instructional technologies" and the ability to "facilitate faculty development workshops, webinars, online courses, small group trainings, etc., in the areas of instructional strategies and emerging technologies." Additionally, these roles

emphasized continuous professional development, requiring candidates to “engage in professional development to stay abreast of emerging trends.”

Figure 9

Required Skills in Higher Education Postings – Comparison of Faculty vs. Director Roles



This data on faculty and director roles highlights the increasing emphasis that higher education instructional design programs are placing on AI as a valuable tool. Currently, instructional designers appear to be acquiring AI skills through on-the-job experience and training as the technology rapidly develops. The data suggests that director-level positions are expected to stay current with industry trends, including having experience in AI and emerging technologies.

Strengths and Limitations

Through this data analysis we have found potential methods for predicting future ID job skills by focusing on the job descriptions of faculty and directors. Based on our analysis, it may be concluded that instructional designers in higher education that aspire to a faculty or managerial role will be expected to have and demonstrate AI skills to be considered for the position. In addition, doing a frequency analysis of job skills requirements and number of overall postings requiring AI or emerging tech skills helps provide an objective measure of what employers currently require.

The data collection and analysis process for this research presents some weaknesses. One significant limitation is the inconsistency in how job postings are presented, making it difficult to ensure a uniform collection process and consistency. While steps were taken to overcome this weakness, each of these solutions also included weaknesses and limitations. ChatGPT can quickly analyze large datasets but its ability to comprehensively interpret job postings is limited due to its reliance on pattern and word recognition. Manual analysis of data is time consuming and introduces additional opportunities for inconsistencies. These challenges underscore the need for more refined methodologies and tools to accurately analyze job postings and capture the desired skill sets and trends related to AI and emerging technologies as they relate to the instructional design field.

Additionally, due to the limited length and time for data collection, any predictive analysis about the number of future job postings requiring AI skills is tentative at best. A much more robust data set would be needed to help solidify the trend line and improve the

value of this analysis. Furthermore, as AI is an emerging skill set there is ambiguity in its presence in job postings; it may be a trend that never makes it into postings for a staff role. We attempted to address this limitation through researcher interpretation of AI being included as an “emerging technology” or similar terms, but that may not be an accurate assumption.

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